

Technical Notes: Reference Management and Study Selection

1. Tool Overview

The systematic literature review utilized **Zotero (Version 8.0.4)** for archival management and bibliography organization, and Rayyan AI for the specialized screening and deduplication phases. The integration of these tools ensured a robust audit trail from initial discovery to final inclusion.

2. Multi-Stage Selection Workflow

The selection process was divided into identification, automated filtration, manual screening, and deduplication:

- **Stage 1 & 2 (Identification & Automated Filtration):** The initial electronic search across five databases yielded **639** records: Web of Science (n=181), Scopus (n=174), ScienceDirect (n=120), IEEE Xplore (n=116), and ACM Digital Library (n=48). Using automated database filters, we restricted the scope to journal papers published between 2021 and 2025, which resulted in the exclusion of **242** records.
- **Stage 3 (Deduplication and Initial Screening):** The remaining records were consolidated into a single RIS file via Zotero and imported into Rayyan AI. Deduplication identified and removed **47 overlapping records**, leaving **350 unique studies** for screening.
- **Stage 4: Title and Abstract Screening** Independent reviewers conducted a title and abstract screening of the 350 records. This phase led to the exclusion of **226 records**, consisting of **14 secondary studies** (surveys and reviews) and **212 records** deemed irrelevant to the technical scope of insider threat preprocessing. This left **124 records** for full-text retrieval.
- **Stage 5: Full-Text Eligibility and Quality Assessment (QA)** Of the 124 records, **7 were excluded** due to paywall restrictions, leaving **117 studies** for deep analysis. After applying the Quality Assessment (QA) criteria, a final total of **110 primary studies** was reached.

3. Justification for Final Exclusions

During the full-text eligibility phase, **7 studies** were excluded for failing to meet the QA threshold. Specifically:

- **Three (3) papers** were removed for lacking sufficient technical documentation regarding their feature engineering pipelines.
- **Four (4) papers** were excluded as their research scope diverged from the specific domain of insider threat detection.

4. Data Integrity and Peer Review

To minimize selection bias, the process was performed in **duplicate**. Two independent reviewers carried out each stage, from title screening to the application of QA criteria. Any discrepancies in inclusion were resolved through systematic note-sharing and collaborative discussion. The complete transparency of this process is visualized in our **PRISMA 2020 Flow Diagram** (Fig. 1 of the manuscript).